BTP **Integration** Capabilities with **Event-Driven Architecture** and **AI**



Barış Büyüktanır

SAP INSIDE TRACK ANKARA 2026

Agenda

1. SAP (BTP) Integration Capability **& SAP BAIP (Business AI Platform)**
2. **Walk through** SAP's Integration and Event-Driven Capabilities, Event-Driven Architecture **"in action"** with Advanced Event Mesh
3. **AI** in Integration Suite
4. **Recap** and **Roadmap** for SAP BTP Integration, AI & AEM
5. Final Words and **Key Takeaways**



SAP Business AI Platform
&
SAP Integration Capability

SAP Business AI Platform

Combined into Single Stack

BTP + BDC + Business AI

Applications

- Financial management
- Spend management
- Supply chain management
- Human capital management
- Customer experience
- Enterprise resource planning

AI Platform

BUILD
- Develop
- Integrate

CONTEXTUALIZE AND REASON
- Data and analytics
- AI models

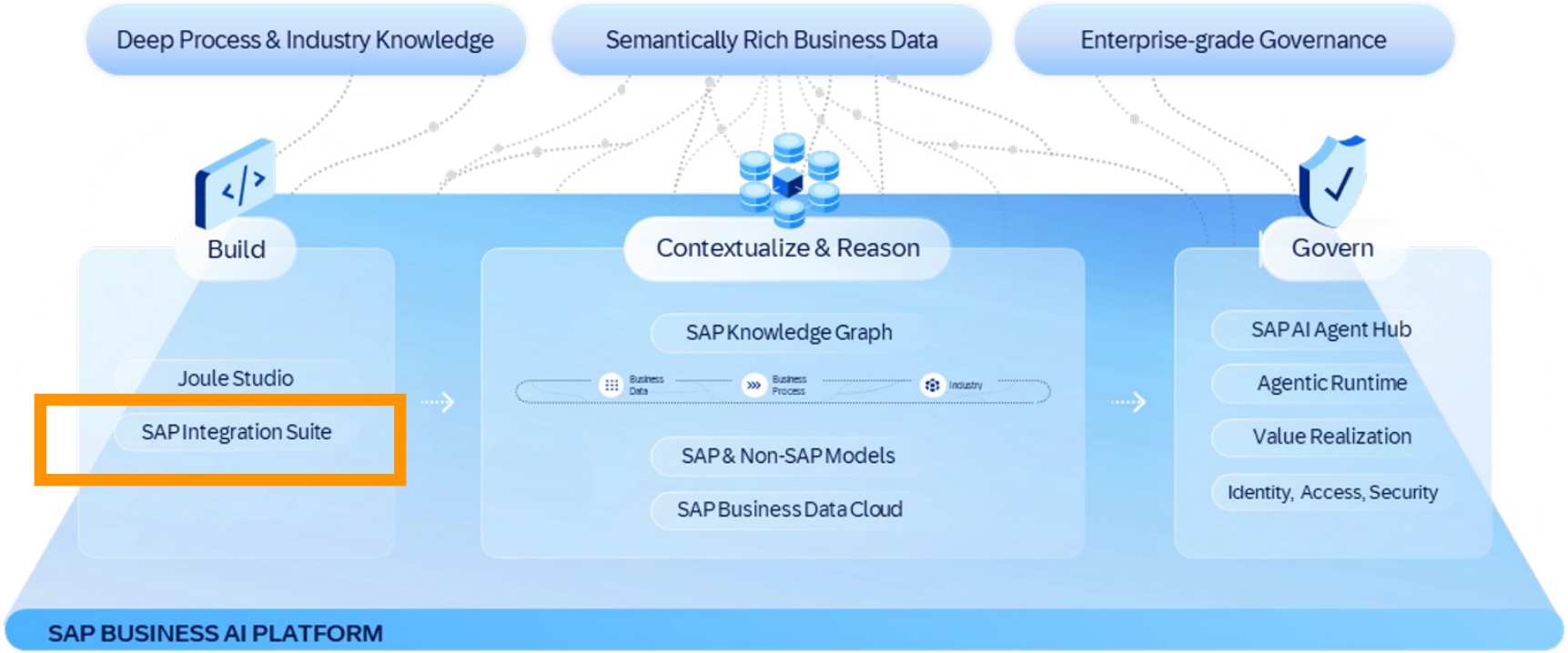
GOVERN
- AI agent governance
- Transformation management

Artificial intelligence

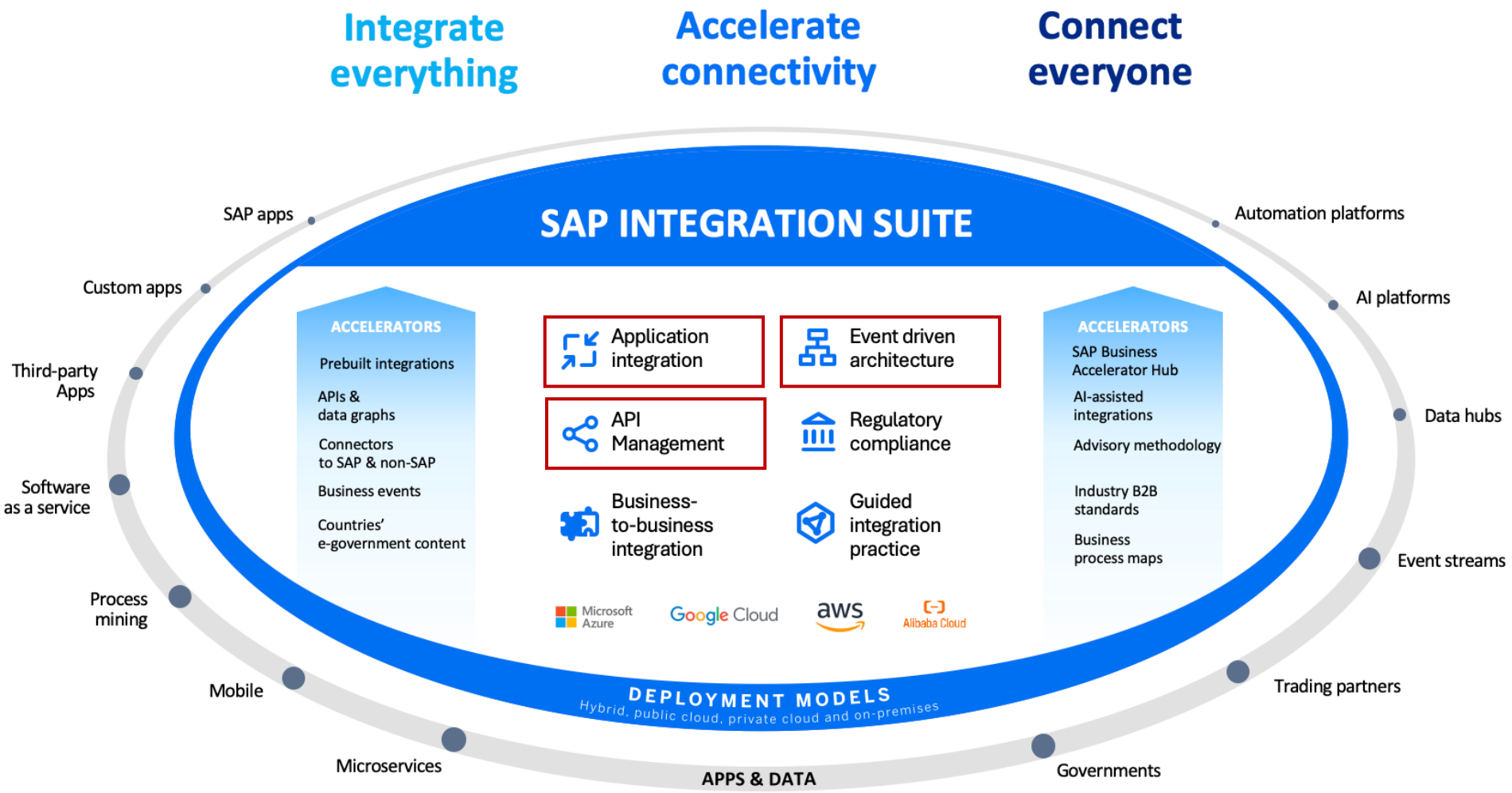
- AI agents and assistants
- Joule Work

More SAP offerings

- RISE with SAP
- Midsize business solutions
- Sustainability management
- Business network
- All products A-Z



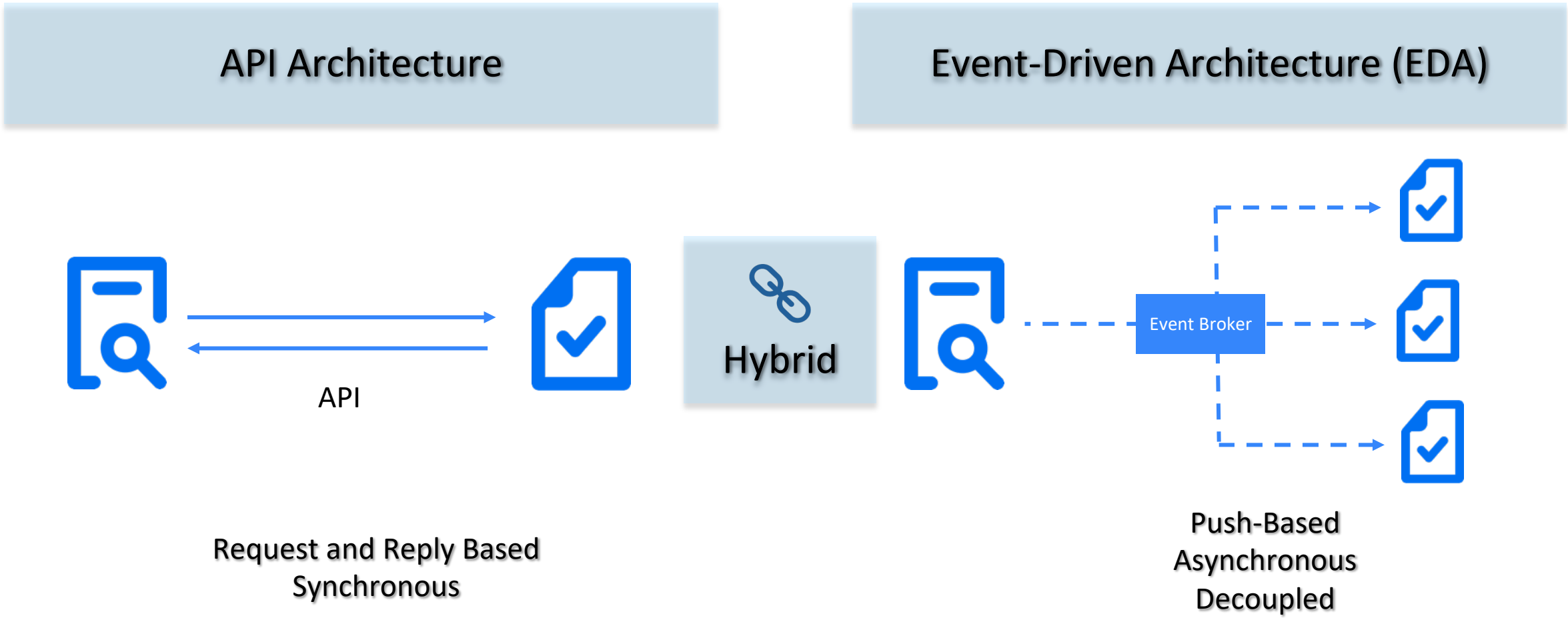
Integration via BTP : **SAP Integration Suite**



Integration Styles

API-led classical, **Event-Driven** and **Hybrid** Integrations

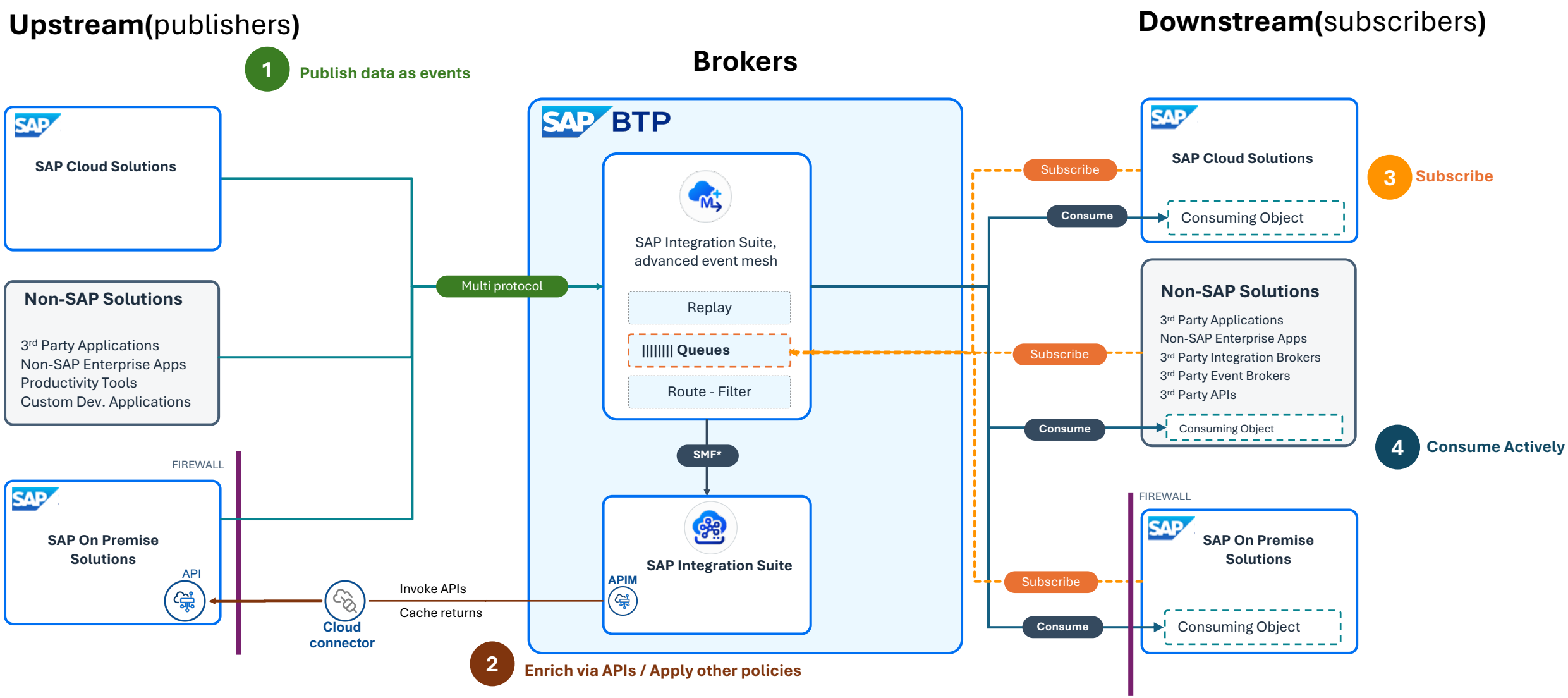
When to use which



Why Event-Driven ? – too many reasons

- Loose coupling of systems and applications (**minimum dependencies**)
- (Near)**real time** information published to **distributed systems..** No polling .. **Everything is event**, every component acts as soon as possible
- **Publish once**, do not stop/wait for other service(s).
- Shifting from **huge batches to small blocks of data** over time
- **Handle slower consumers**, back-off / back-pressure
- Load balance - **scale and fail independently**, add new consumers easily
- No loss of messages, **replay the history** for late joiners / errors
- **«Smart routing»** based on topics

The Big Picture – Architecture



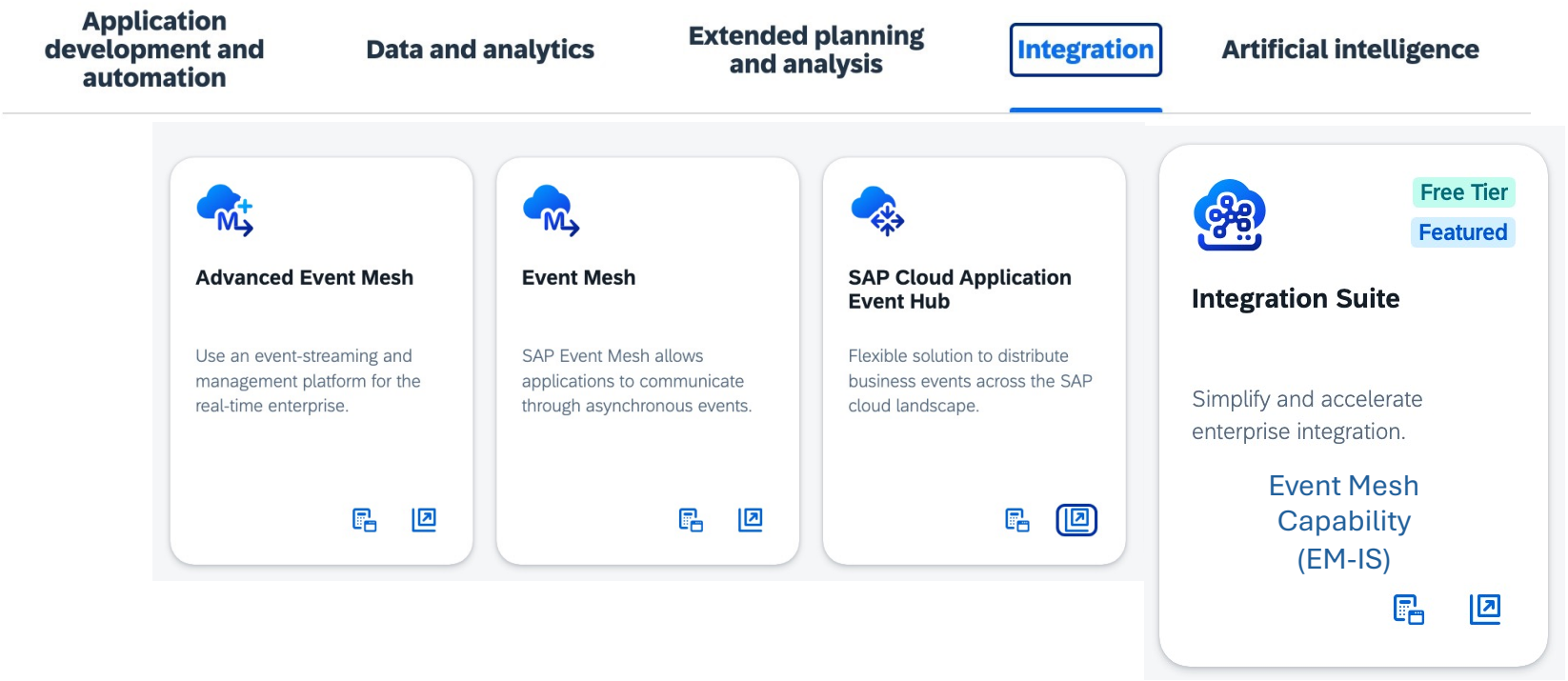


Walk Through **SAP's Integration & Event-Driven** Capabilities

SAP’s Event Brokers

SAP **Event-Driven Integration Offerings** for BTP

EMIS-first, AEM-for-scale, standalone Event Mesh-as-legacy



Advanced Event Mesh

A lot of **Event-Driven Use Cases**

HR across multiple systems	Data Lake Updates	Medication Delivery and Tracking	Flight Tracking	Production Line Analysis	Demand Based Supply Chain	Attendance Tracking	Financial Market Data Distribution	Global Event Localization and Distribution	Inventory Management	Event-Driven Forecasting
Weather based Pricing	Water Quality Monitoring	Harbour Management	Elevator Emergency Service	Real Time Inventory	Sales Support	Budgeting and Forecasting	Smart Order Routing	Meter to Cash	Ticketing	Document Update Notification
Supply Chain Control Tower	Digital Twin	Supply Chain Visibility	Customer Guidance	Shipment Notification and Tracking	Credit Card Authorizations	Automatic Ordering	Food Quality Tracking	Sales Order Shock Absorber	Oil Well Monitoring	Mobile Workforce Connectivity
Industrial Maintenance	Real Time Trading	Real Time Loyalty Points	Luggage Tracking	Buffering when Offline	Smart Financial Order Routing	Across Vendor Business Processes	Regulatory Reporting	Online Store Buffering	Smart City	Event Log / Kafka Data Updates
Smart Meter Readings	Predictive Maintenance	Real Time Order Management	Connected Car	Real Time Test Data	Across Vendor Integration	Airport Management	Real Time Payments	Point of Sale	Smart Trashcans	Business Process Documentation

Smart Routing – multi source / multi target

TOPIC [object] / [action] / [source] / [objectID] / [timestamp] / [type]

Events published

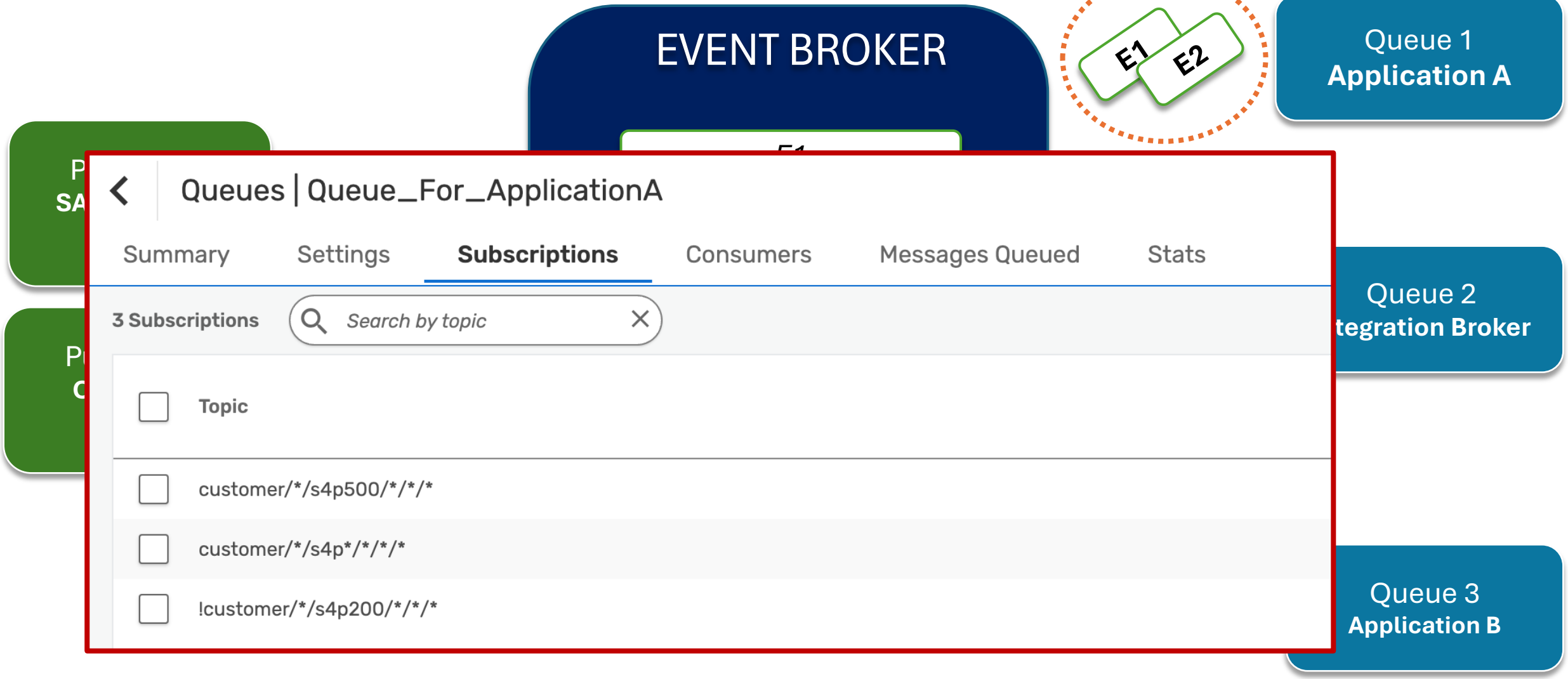
EVENT 1	customer	/	created	/	s4p500	/	998844	/	20250510131522	/	XYZ
EVENT 2	customer	/	created	/	erp200	/	999423	/	20250510131522	/	ABC
EVENT 3	customer	/	changed	/	s4p500	/	998844	/	20250510134322	/	ABC

Route Events via Filters

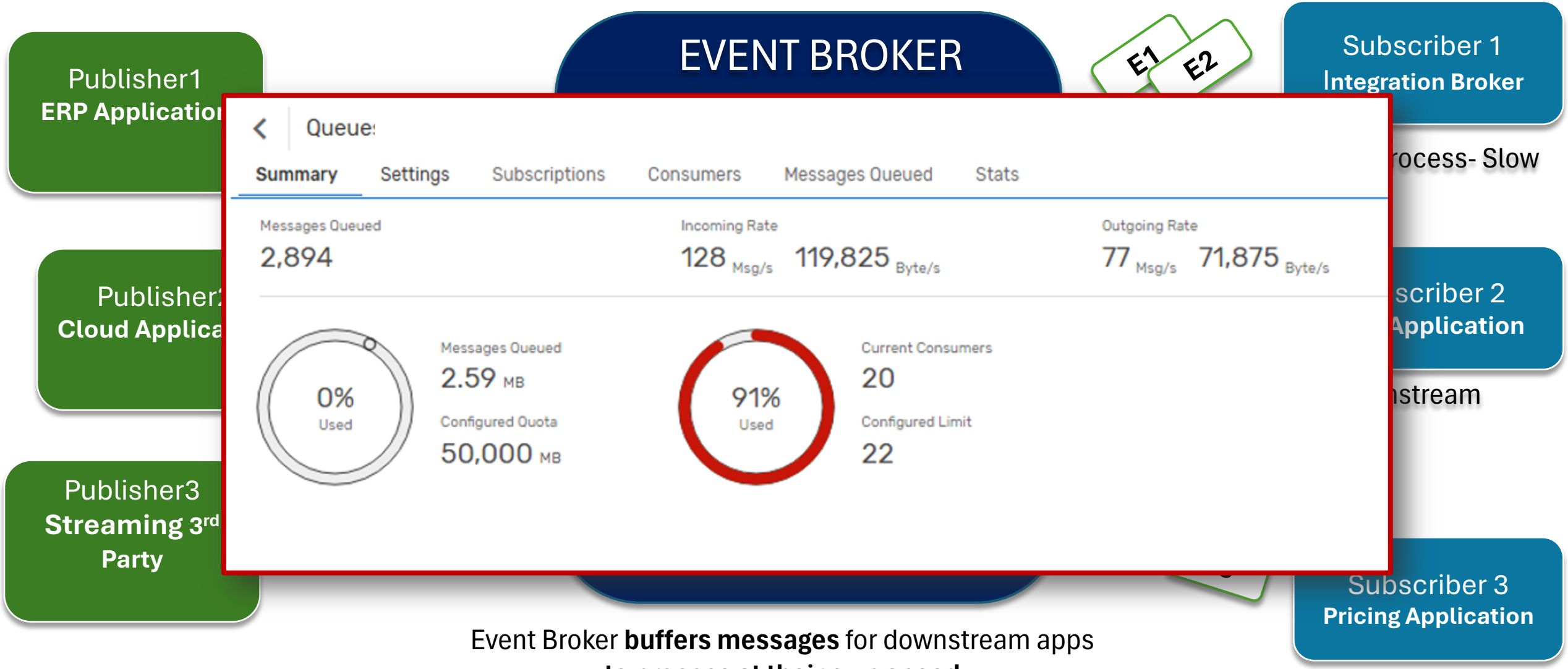
FILTER 1	customer	*	s4p500	*	*	*
FILTER 2	customer	created	*	*	*	XYZ
FILTER 3	*	*	*	998844	*	*

Smart Routing – multi source / multi target

Millions of Events ..MANY "loosely coupled" sources and targets



Managing back-pressure – Relaxing Consumers



Event Broker **buffers messages** for downstream apps
to process at their <u>own speed</u>

SAP API Management

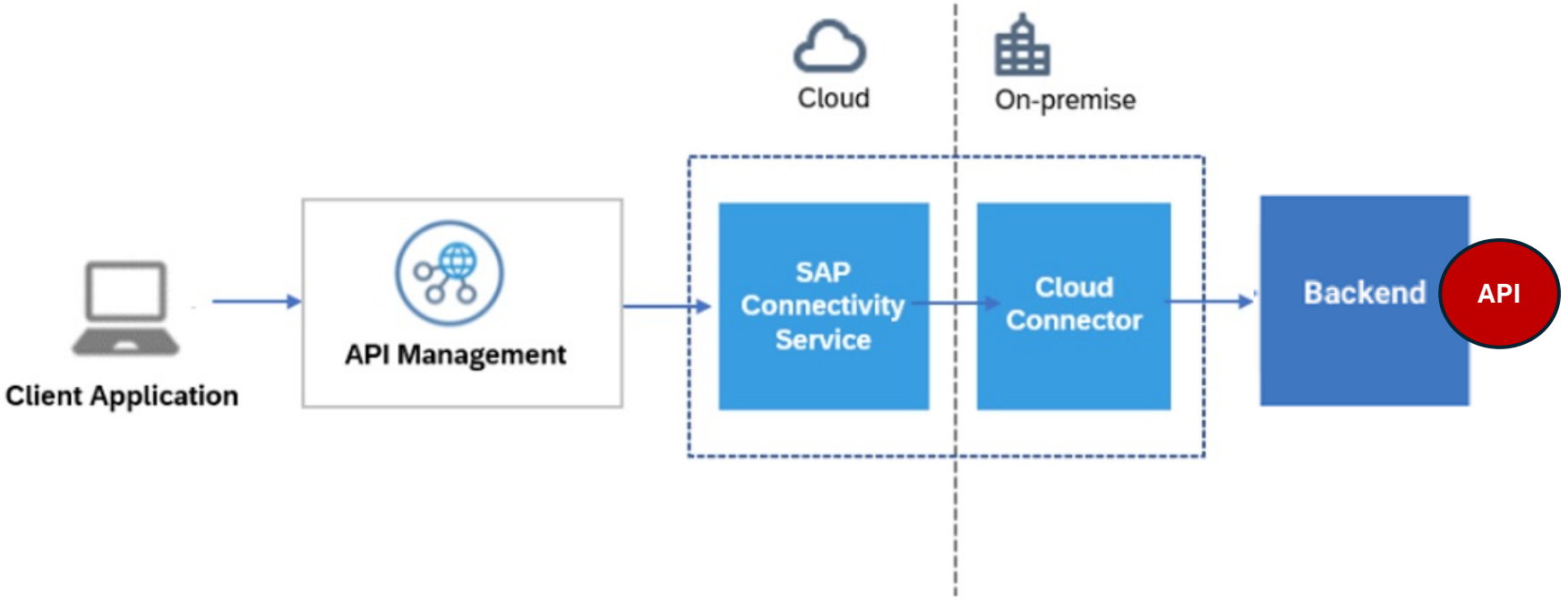
“APIs to be reachable»

What is API Management?

in a
Secure
Scalable
Managable
way

Expose
APIs (services)

From
backend systems



Real-World Use Cases of SAP API Management

Response Caching
Improve performance by caching repetitive requests when backend data changes infrequently.

Key-Value Map (KVM)
Manage dynamic parameters like tokens or endpoints externally using get/set scripting.

Quota + Spike Arrest (Rate Limiting)
Protect public APIs from overload and define per-subscriber call limits.

VerifyAPIKey + OAuth2 Security
Control who accesses your API with robust, app-based authentication policies.

Backend Failover / Mock Response
Ensure business continuity with static or KVM-based fallback responses when backend fails.

JavaScript Payload Manipulation
Enrich responses with dynamic fields like timestamp, clientId, or geo info.

Header Injection & Sanitation
Auto-inject headers (e.g., Authorization, X-Correlation-ID) to enhance security and traceability.

Request/Response Validation (JSON/XML)
Enforce strict schema validation to prevent malformed data from affecting backend systems.

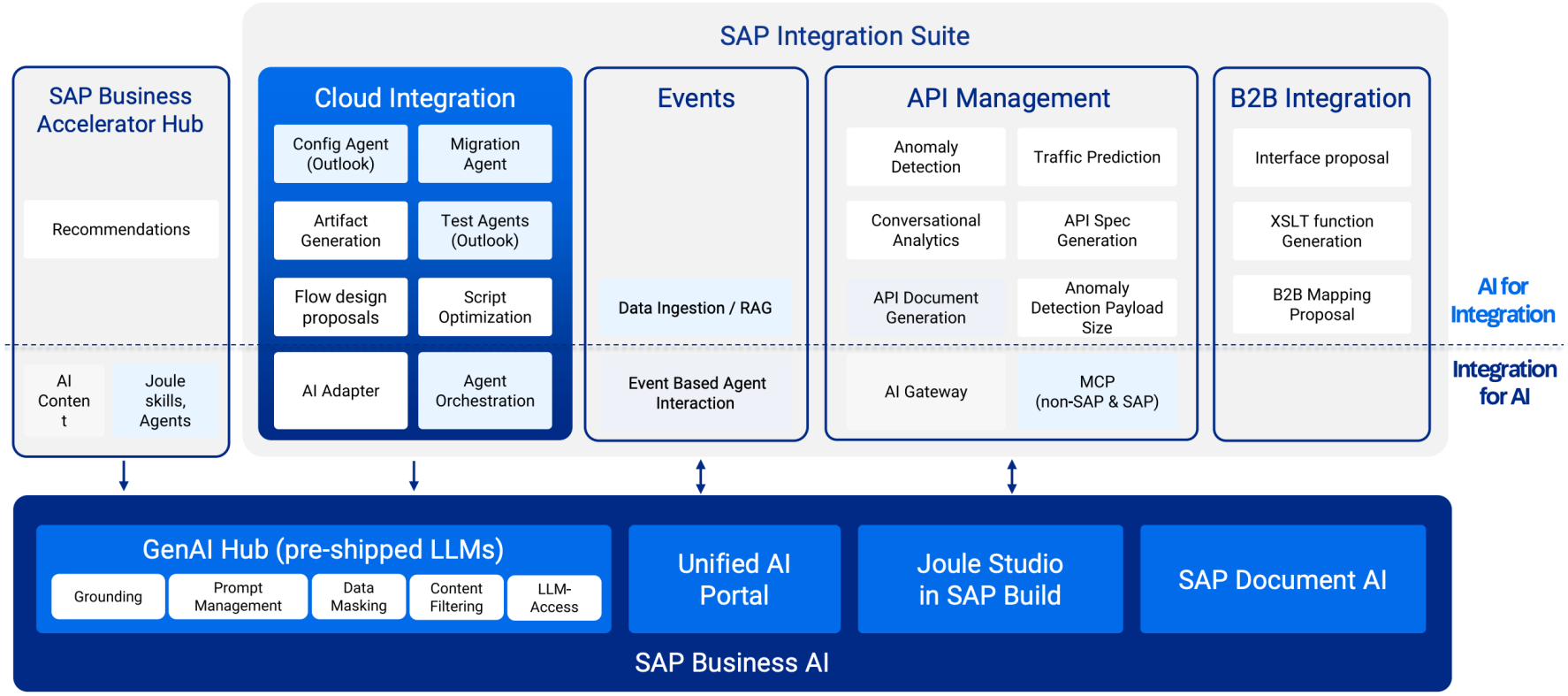
AI
in
Integration Suite



AI in Integration Suite

AI for Integration

Integration for AI



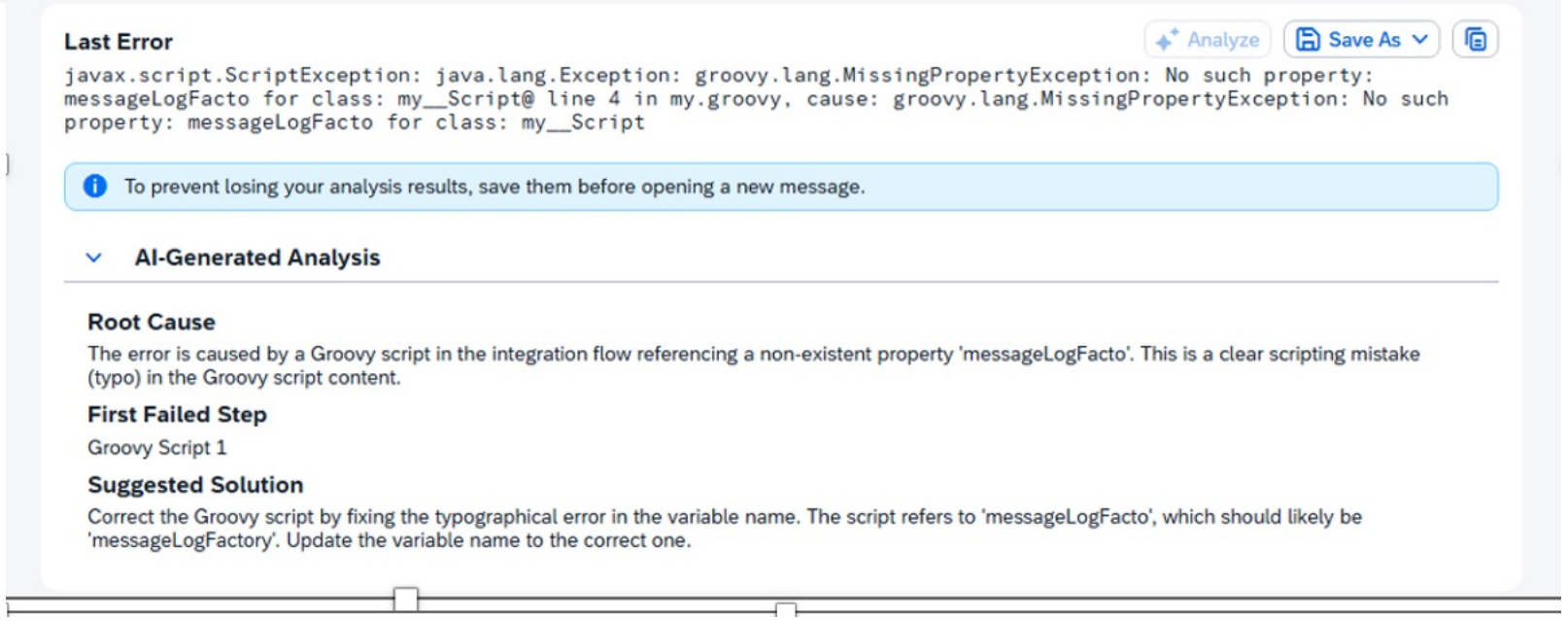
AI for Integration

AI for Integration

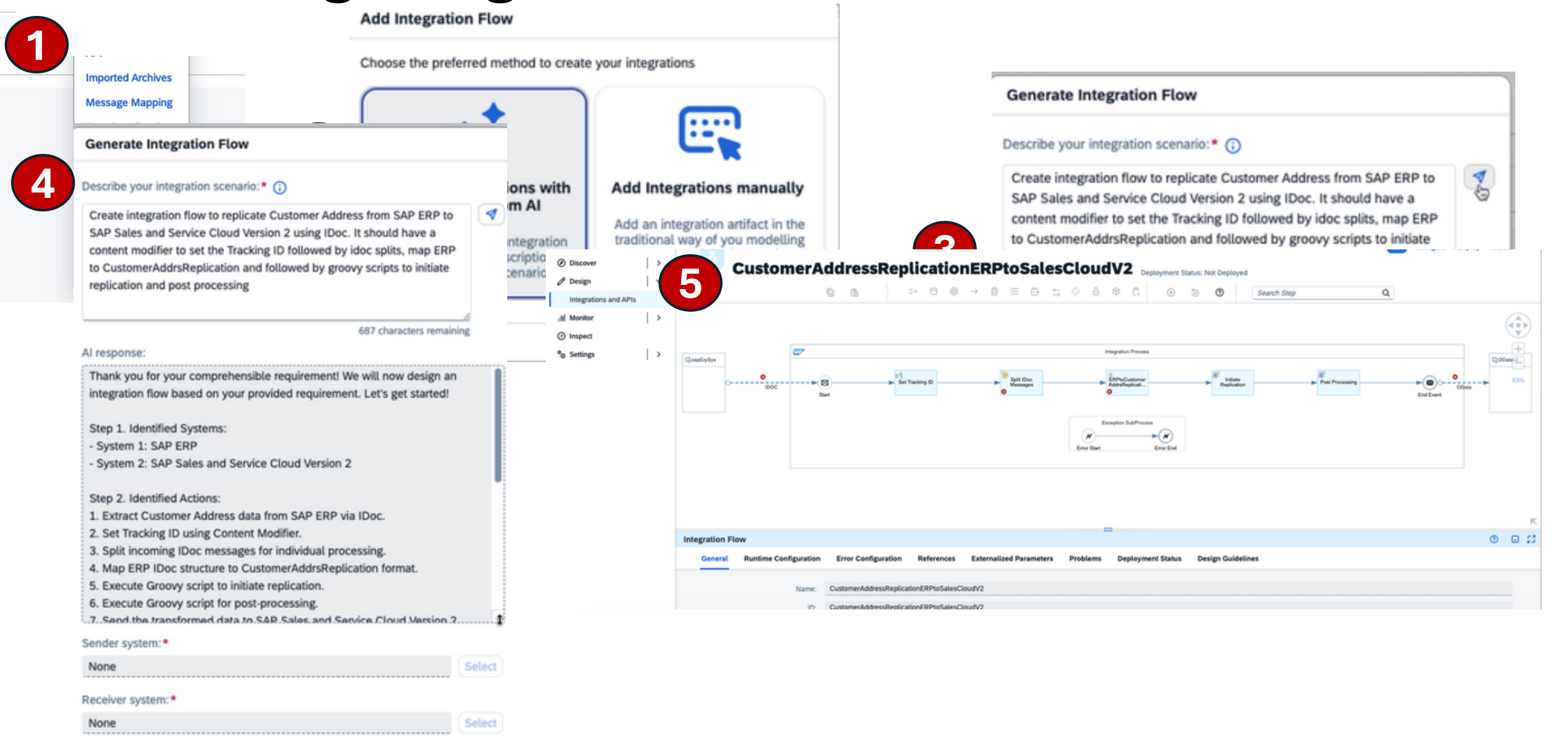
Feature	Capability	Description
AI-assisted Error Resolution	**Cloud Integration**	Use artificial intelligence to analyze and resolve message processing errors.
Flow Step Recommendations	**Cloud Integration**	AI recommends the next flow steps, while adding flow steps in your integration flow to help simplify the design process.
Generation of Integrations	**Cloud Integration**	Upon activation, you can create integration flows with assistance from Generative AI.
Script Optimization	**Cloud Integration**	Upon activation, enhance your Groovy scripts using Generative AI.
Memory Profiling	**Cloud Integration**	Shows the memory insights derived during integration flow simulation and suggests improvements.
Integration Content Migration Agent	**Cloud Integration**	Converts Java mappings to Groovy scripts during the migration of SAP Process Integration and Process Orchestration scenarios to integration flows.
Anomaly Detection	**API Management**	Upon activation, performs system-level checks in the background to identify deviations in API call patterns from expected behaviour.
API Traffic Prediction	**API Management**	Upon activation, identifies API traffic trends and provides predictions for upcoming call volumes.
Generation of OpenAPI Specification	**API Management**	Upon activation, you can leverage AI assistance to automatically generate an OpenAPI Specification while creating an API artifact.
MIG & MAG Proposals	**Integration Advisor**	Benefit from message detail proposals for MIGs and mapping entity proposals for MAGs. Upon deactivation of this service, MIG and MAG proposals will not be accessible to the tenant, and data won't be collected from this tenant anymore (although this will only impact future data collection, leaving previously collected data unaffected).

AI assisted error resolution



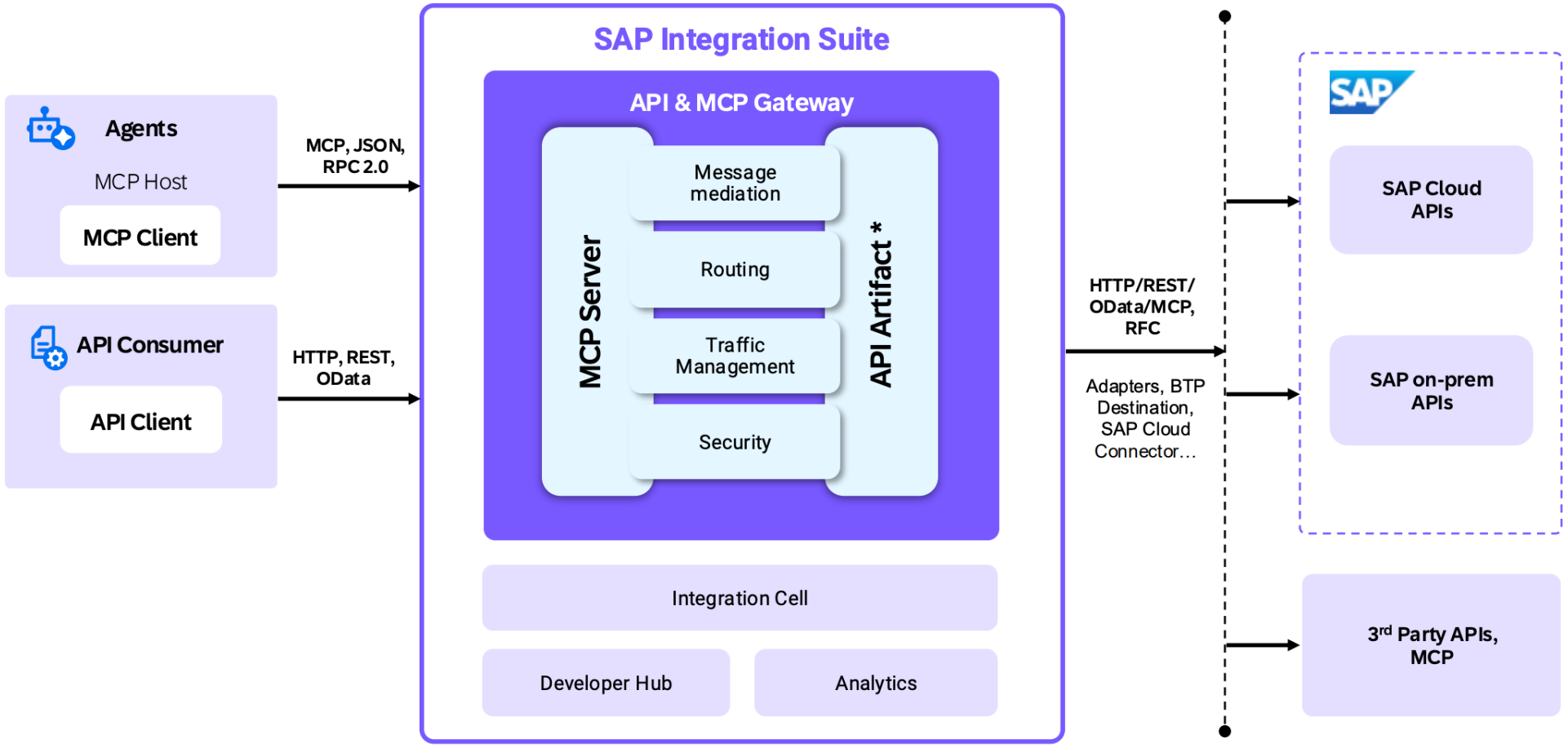


Generating Integration Flows with AI Assistance

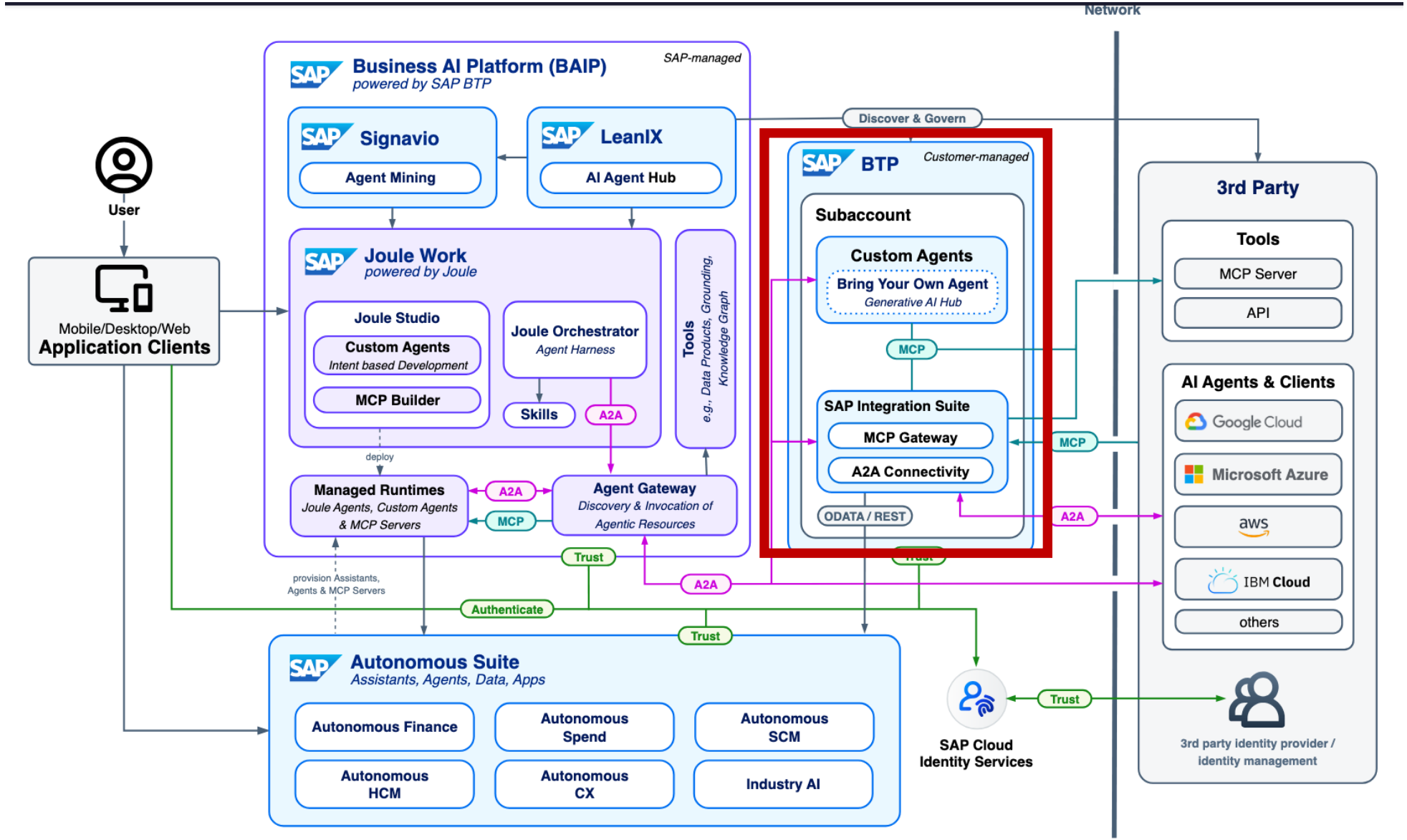


Integration for **AI**

Integration for **AI -** The Protocols – Agents, A2A, MCP



The Architecture– AI Capabilities



Integration for **AI** – example flow

Backend

Integration Suite

AI Agent

API

Develop and Expose API

API Proxy

Configure Access to API

MCP Gateway

Configure MCP

Connect & Talk (MCP)

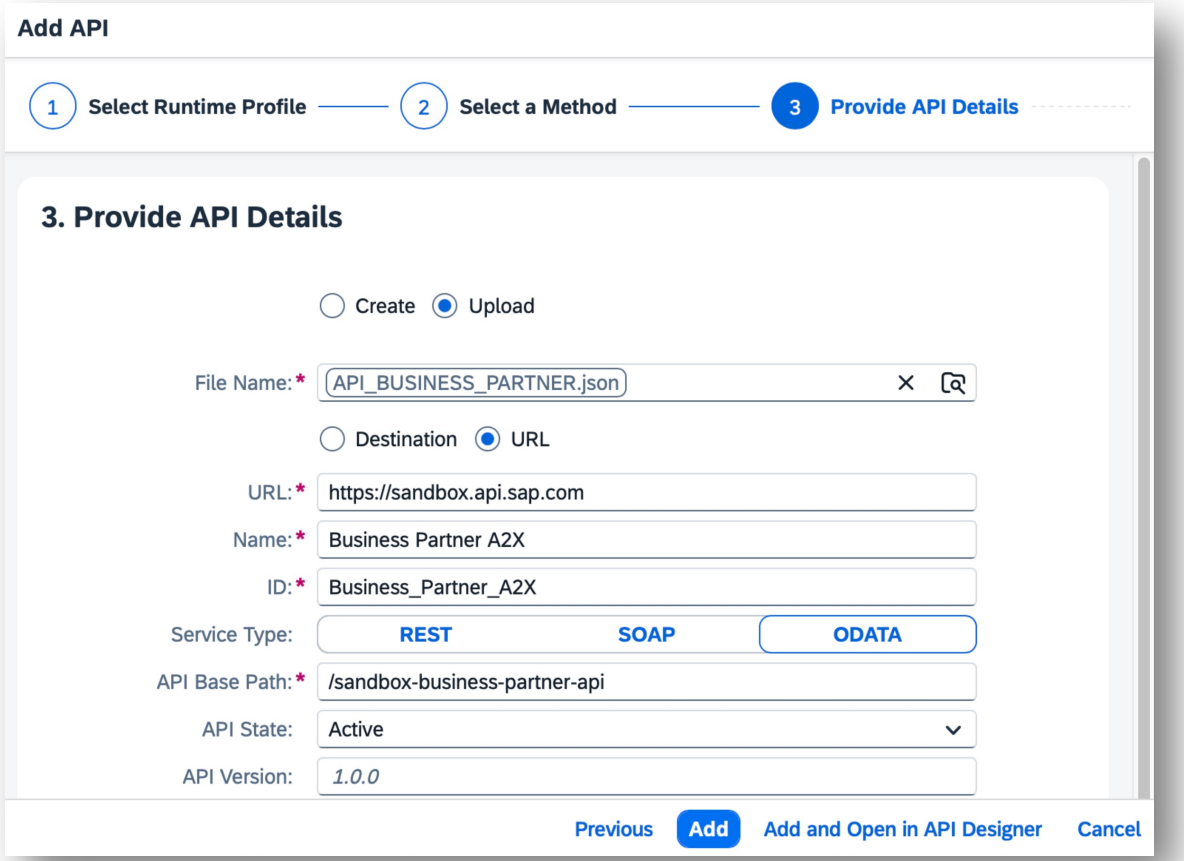
Configure Access to MCP Gateway

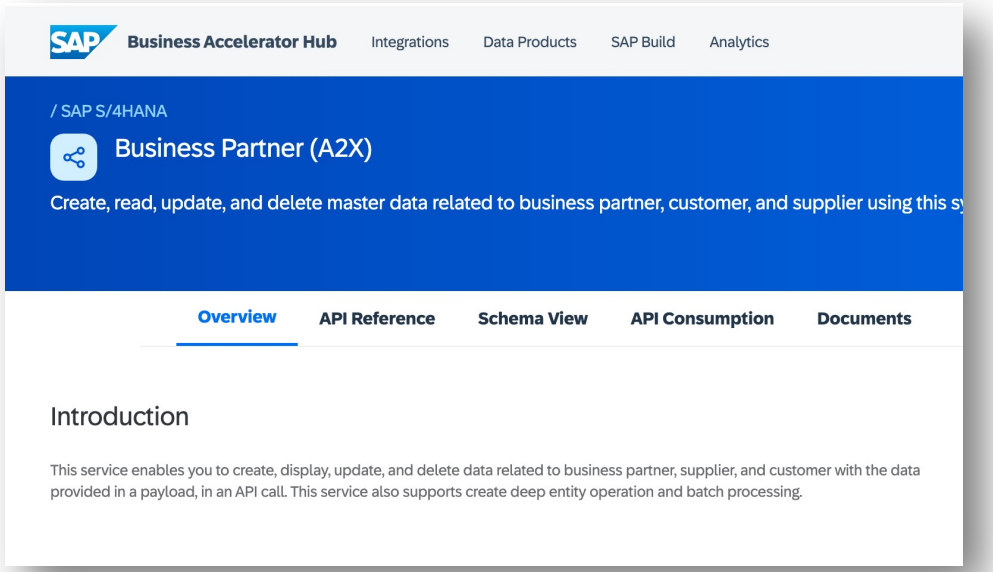
Ask for tasks

Integration for **AI** – example flow

Integration Suite

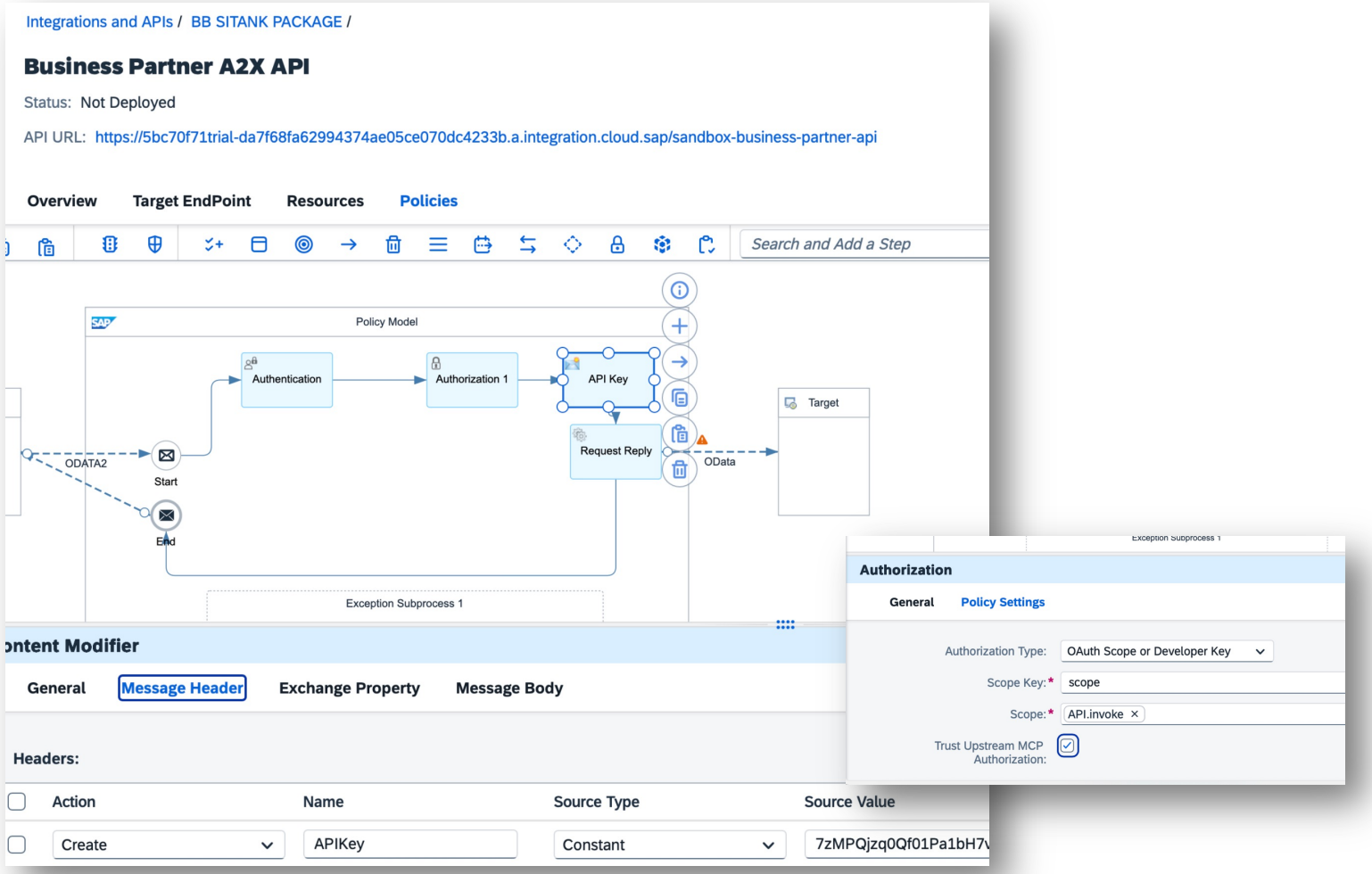
Backend



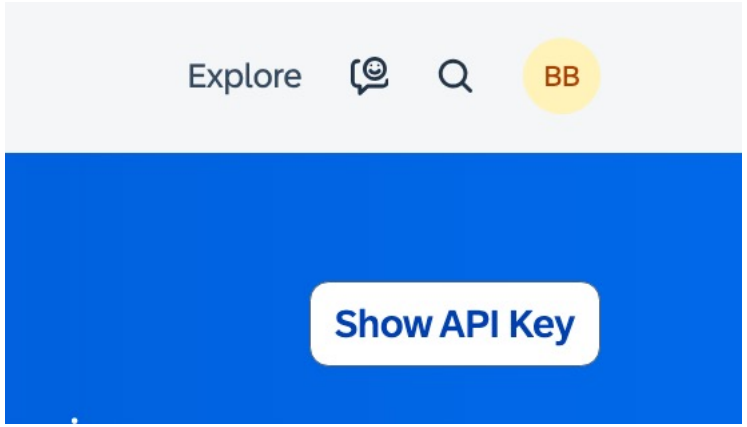


Integration for **AI** – example flow

Integration Suite

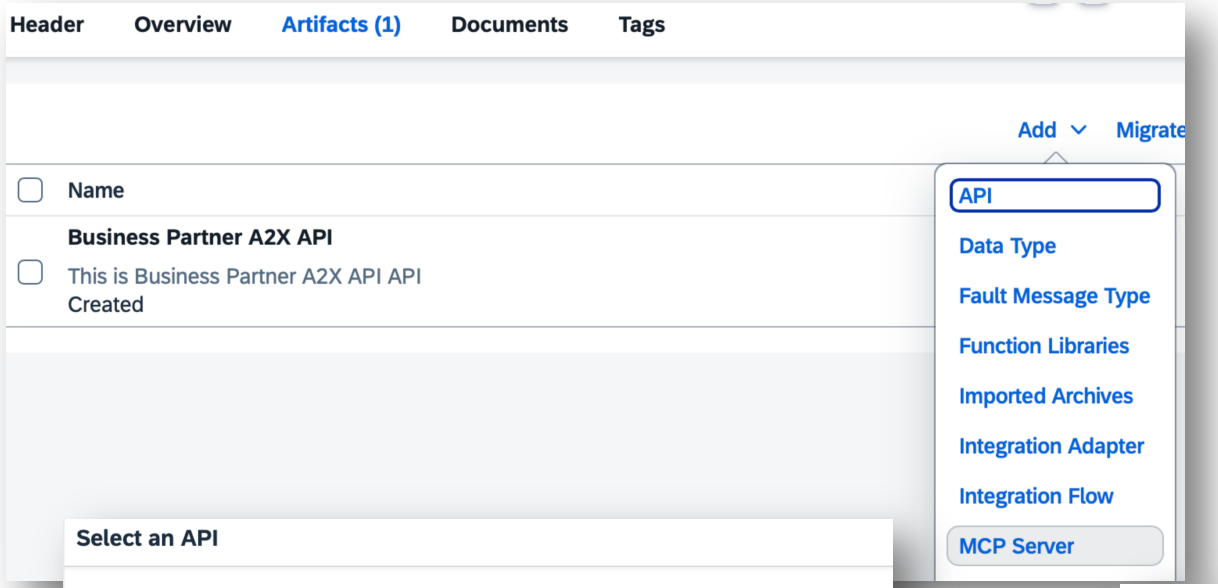


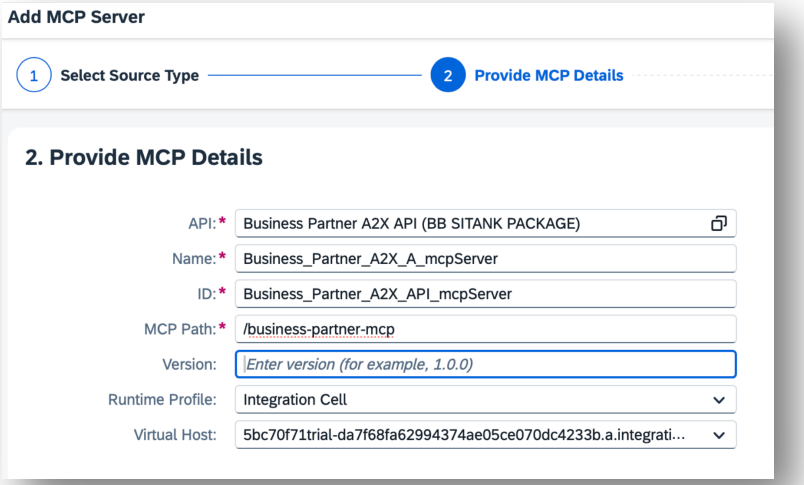
Backend

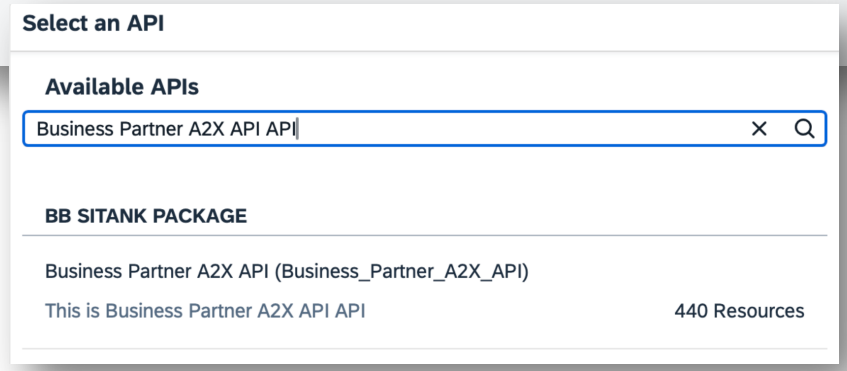


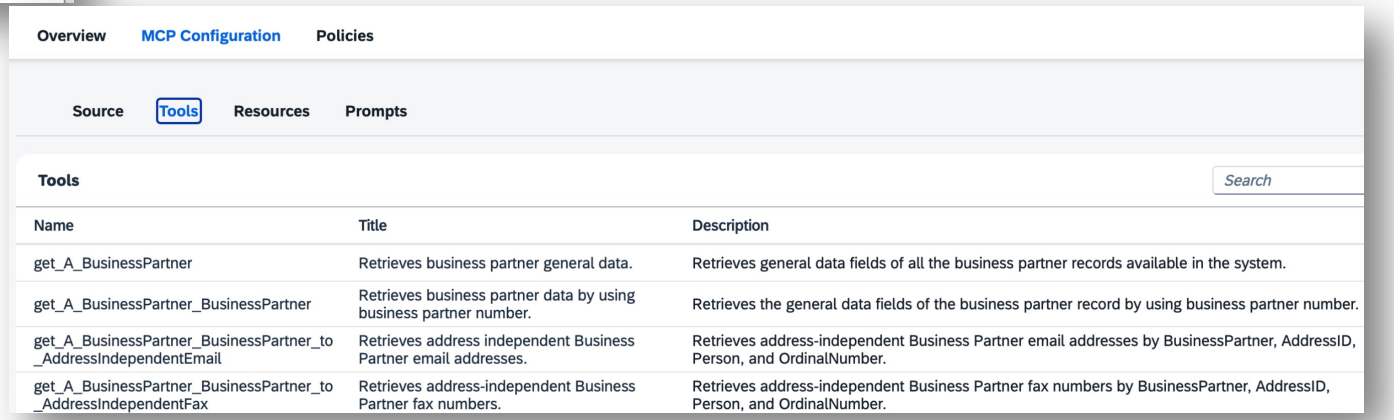
Integration for **AI** – example flow

Integration Suite



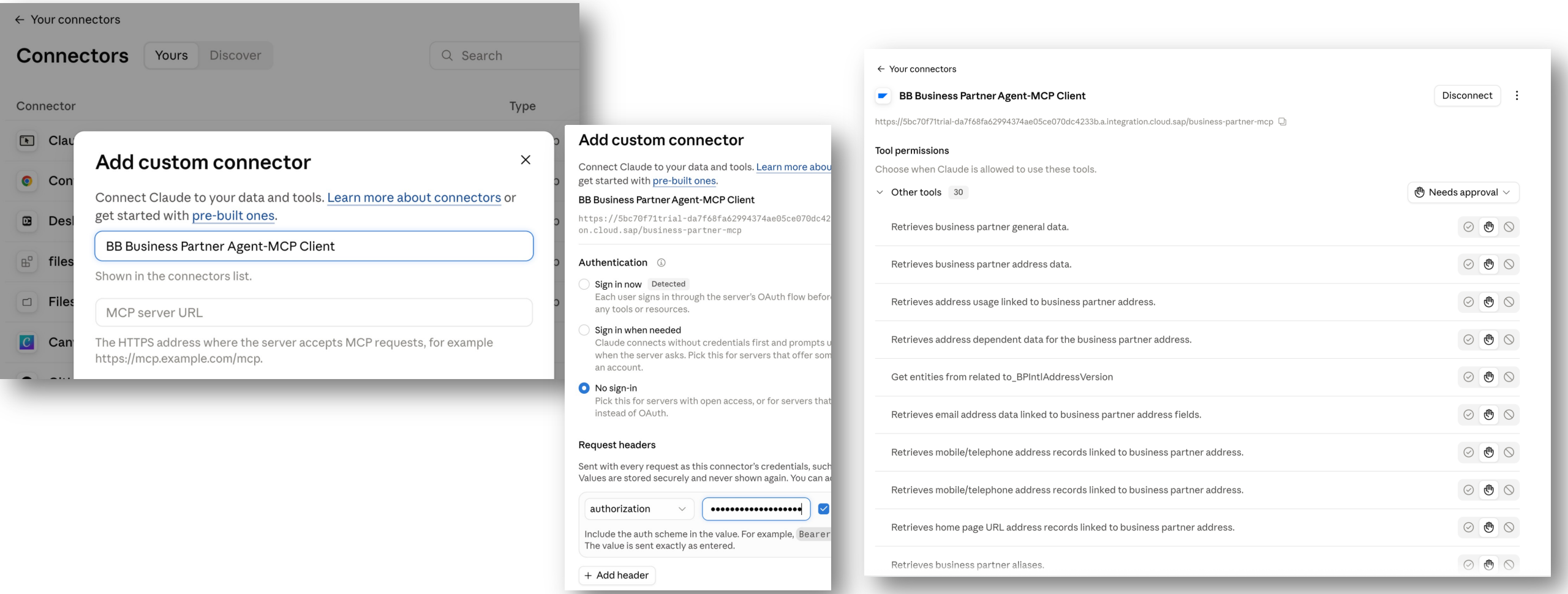






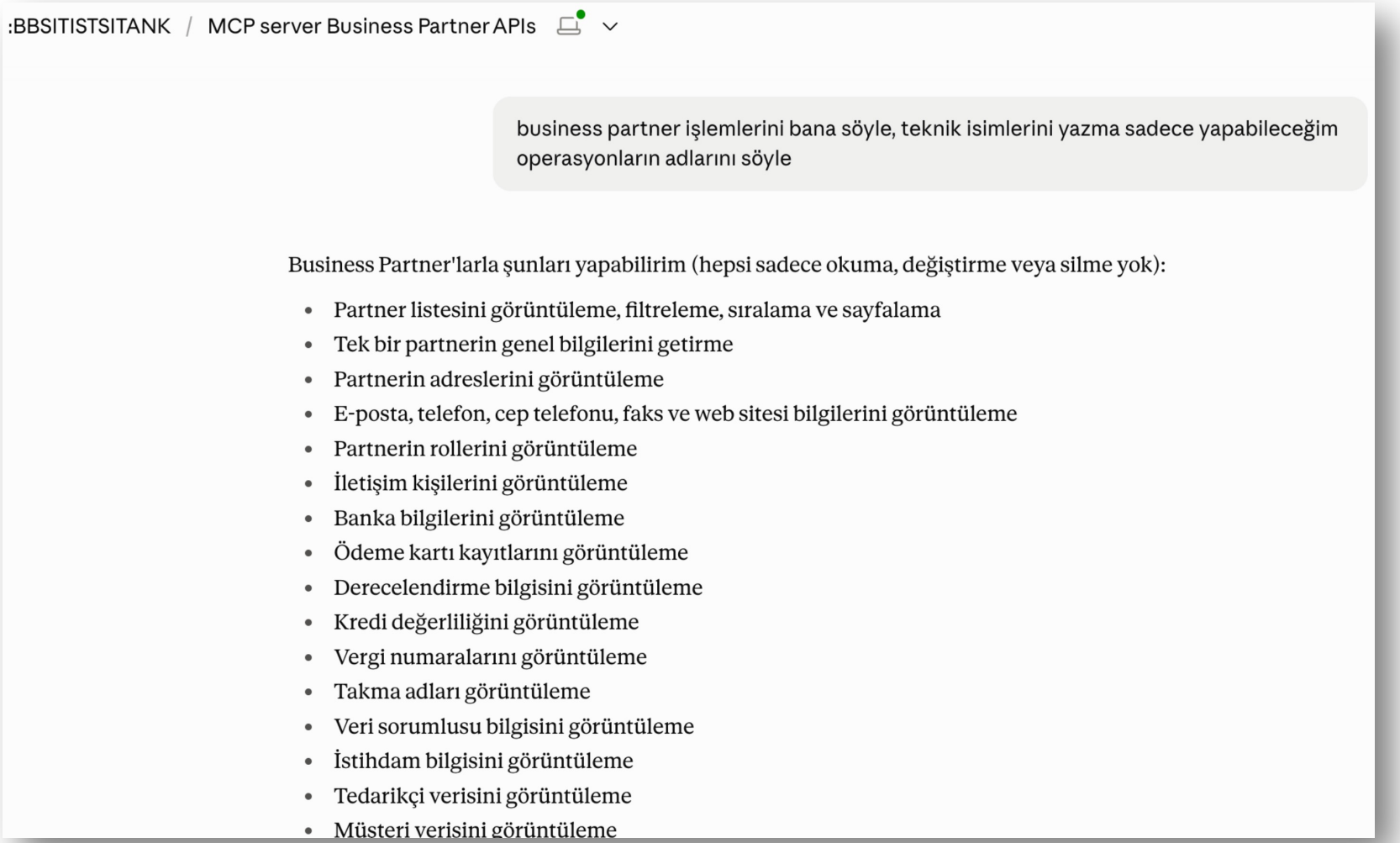
Integration for **AI** – example flow

AI Agent



Integration for **AI** – example flow

AI Agent



Integration for **AI** – example flow

AI Agent

Banka bilgileri olan 5 tane business partner listele ve yanlarına bankalarını yaz.

Retrieving business partner bank details.

Önce banka kayıtları aracının şemasını yüklüyorum.

Used BB Business Partner Agent-MCP Client >

Banka kaydı olan ilk 5 Business Partner (numaraya göre sıralı):

BusinessPartner	Ad	Banka	Ülke	Şehir
1000140	Cust10	CITIBANK N.A.	US	New York
1000170	Global Relief Foundation Inc	HSBC	GB	London
1000171	PSG Holdings LLC	ABN AMRO BANK N.V. - SAMPLE BANK	US	New York
1000290	Inland Corp	Bank of America - NY	US	Palo Alto
1000291	Power Sports Inc-N	Bank of America - NY	US	Palo Alto

Integration for **AI** – example flow

AI Agent

Seçeceğin 5 tane BP için kredibilite listele. varsa yanlarına vergi numaralarını ve banka isimlerini ekle ve tablo yap

Checking the business partner's credit worthiness details.

Önce banka kaydı olan 5 partnerde kredibilite, vergi ve banka verisini tek sorguda çekiyorum.

Used BB Business Partner Agent-MCP Client: Retrieves business partner general data. ›

Bu 5 partnerde kredibilite ve vergi kaydı yok (`null` ve boş dönüyor). Kredibilite kaydı olan partnerleri bulmak için filtreyle arıyorum.

Used BB Business Partner Agent-MCP Client: Retrieves business partner general data. ›

Kredibilite kaydı olan ilk 5 Business Partner:

BusinessPartner	Ad	Kredi derecesi	İflas	Hukuki süreç	Vergi no	Banka
1000000	BANK5115	A	Hayır	0	-	-
1000001	BANK ICS	A	Hayır	0	-	-
1000010	Bank5	A	Hayır	0	-	-
1000162	Bank of America	A	Hayır	0	-	-
1000164	Citibank	A	Hayır	0	-	-

- **Kredibilite:** Beşinde de kredi derecesi `A`. Kredi kuruluşu, standing durumu, tarih ve yorum alanları boş. İflas ve icra işareti yok.
- **Vergi numaraları:** Bu partnerlerin hiçbirinde partner seviyesinde vergi numarası kaydı yok. Müşteri veya tedarikçi kaydındaki vergi numarası alanlarına henüz bakmadım. İsterseniz onları da sorgulayabilirim.



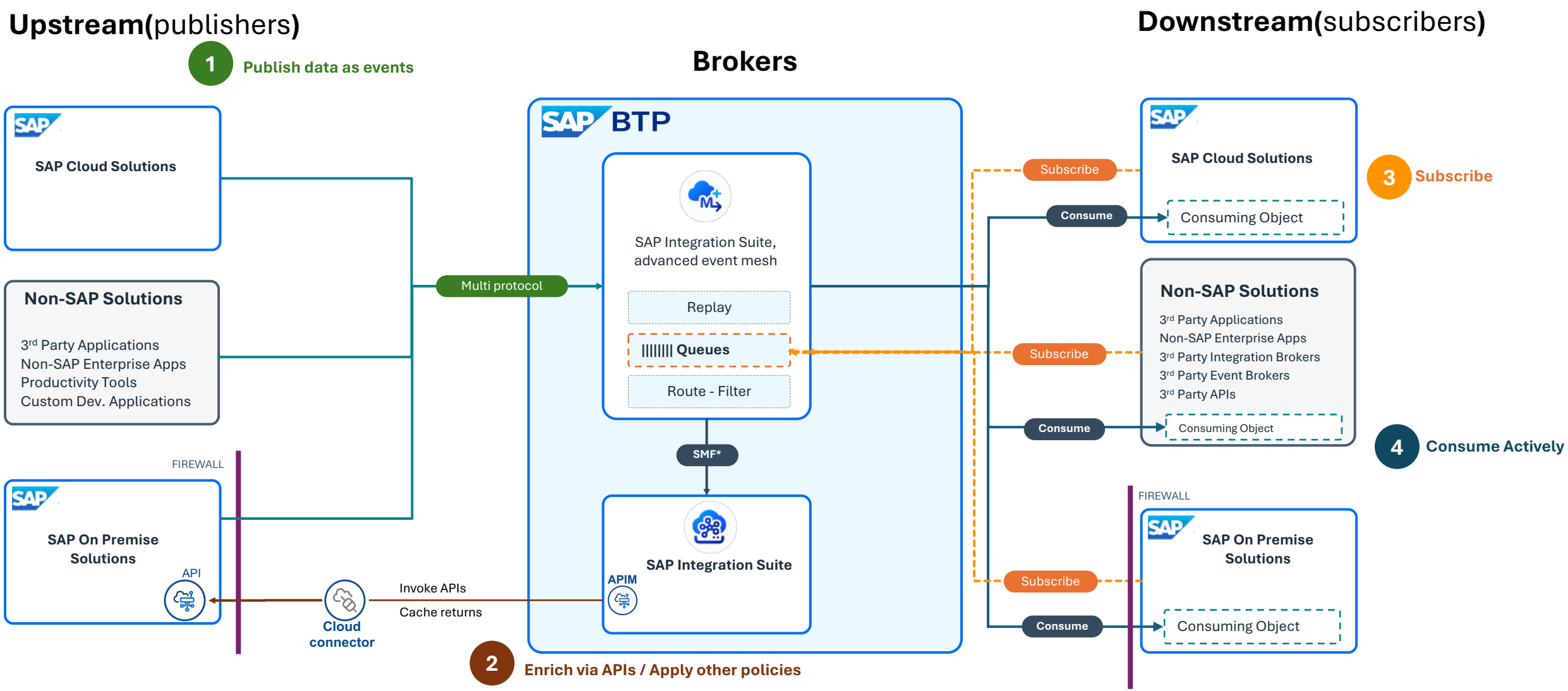
Recap & Next..

” Integration-AI & EDA Roadmap and Take Aways ”

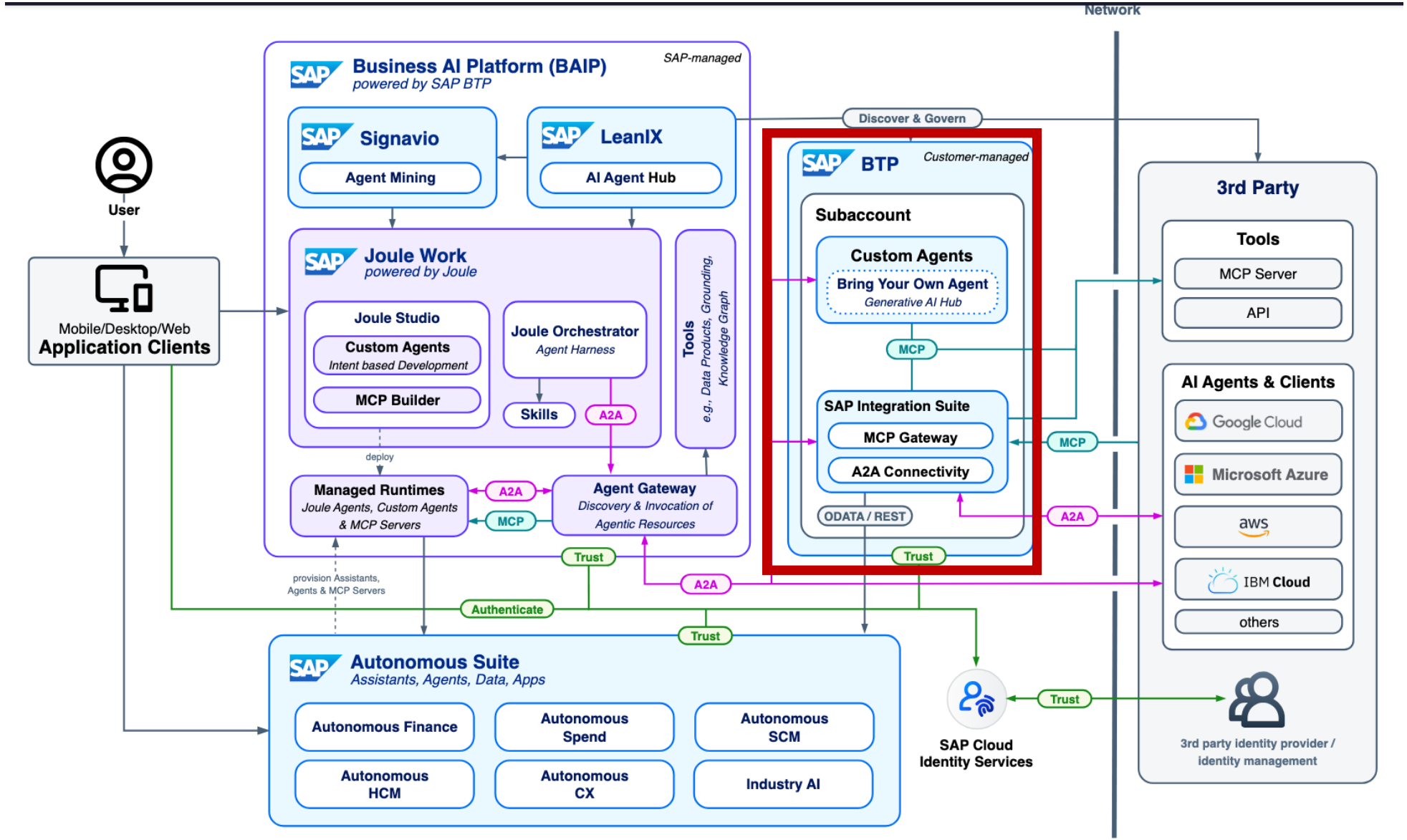
Recap & Next..

” Integration & EDA Roadmap and Take Aways»

The Big Picture – Architecture



The Architecture– AI Capabilities



The Roadmap – BTP Integration, AI & Advanced Event Mesh

Q4 2026

FUTURE RELEASE

Expose integration artifacts as MCP servers

FUTURE RELEASE

Expose remote MCP server as hosted MCP servers in Integration Suite for...

FUTURE RELEASE

Integrating Joule to leverage AI features

FUTURE RELEASE

Distributed tracing enablement between SAP S/4HANA and advance...

FUTURE RELEASE

Joule-powered Event Mesh artifact generation

Q1 2027

FUTURE RELEASE

Advanced event mesh ERP add-on event (Async API) publication

FUTURE RELEASE

AI-based payload size anomaly identification in API traffic

FUTURE RELEASE

External Model Context Protocol (MCP) servers as part of Developer...

FUTURE RELEASE

Model Context Protocol (MCP) server verification in SAP AI Agent Hub

Roadmap URL

Final Words - Key Takeaways

For AI

- **Adopt Agents & AI deliberately** – landscapes are already complex, don't add to it
- Consider the governance, **control**, audit and **security**.

For your integration architecture

- Design for the **long-term** benefits and think **end-to-end** (consider maintenance, reusability)
- **Modernize** your existing interfaces whenever possible (replace inefficient, old-technology)
- Always **start smart & small**

Thank you..



Barış BÜYÜKTANIR

baris@blackbelt.solutions

https://www.linkedin.com/in/barisbt/

@barisbt